[Paper Critics 2] Recent Progress on 3D NAND Flash Technologies

1. Paper Overview

   This paper is written by a senior technical expert at Micron Memory Japan and launched in 2021《Electronics》. At that time, 3D Flash Memory was at a critical juncture, marking a breakthrough from the early 24-layer model to 176 layers and achieving an areal density exceeding 10 Gb/mm^2. This review paper traces the core of the physical limitations on 3D NAND for breakthrough from traditional 2D planes. It goes through in 3 dimensions to construct the overview structure. First, physical scaling, focusing on the geometric evolution in XYZ three dimensions. Including vertical layer stacking (Z-stacking), Z-shrink (character line spacing compression) and layout improvement of memory holes on the horizontal plane to approach hexagonal densest packing (HCP). Second, discussing history of storing more bits per cell in single physical unit, analyzing the trending of TLC becoming the mainstream performance and QLC moving towards affordable high density. Then evaluating the experimental explorations for QLC (such as PLC and low-temperature HLC). The third is circuit and architecture. This study explores the driving force of array-based CMOS technology on die size reduction and write parallelism and explains the write and erase control mechanisms to address the short-term data retention defects and block size scaling in charge-trap cells.

2. Problem Definition

   According to the flash memory structure from 2D to 176-layer high density 3D, the base physical restrictions and system integrations meet multiple bottlenecks: asymmetric operation delay and microscopic leakage mechanism, high aspect ratio etching and geometric limits and erasing granularity out-of-control cause the system burden.

   Asymmetric operation delay and microscopic leakage mechanism, in the architecture of RG-CTN, it needs to erase data by BE-ONO to DT and write by high-voltage electronic tunneling. However, SiN storage layer causes very serious lateral migration and vertical relaxation by its continuous architecture. In addition, the charge at the entrance of the tunnel layer rapidly detrapping. It leads to the critical voltage of the latest written unit occurring unexpectedly sink and widen in some seconds or minutes. It makes it needs to accurately divide to 16 statuses for QLC writing time to fulfill 4 or 5 times of TLC.

   High aspect ratio etching and geometric limits. Vertical stacking to 176 layers causes the aspect ratio increase rapidly. If the space between character lines is not compressed synchronously, the memory pillars are very prone to bending or bottom closure during chemical plasma etching. However, thin interlayer insulation can exacerbate leakage between upper and lower cells, forcing the

industry to evaluate highly costly disruptive methods such as vertically halving the column or physically isolating SiN.

Erasing granularity out-of-control causes the system burden. In 3D architecture, character lines are distributed in a whole plate shape, it makes that the single erase block size would increase to MB even GB level by the increasing of layers. Flash memory has a physical property of writing in page and erasing in block, leading to it must carry many valid data in garbage collection. It would cause WAF to increase extremely and critical tail latency spikes.

3. Methodology

   The review paper is based on the real chips data launched by IEEE ISSCC, comparing the density of area plane, bandwidth of writing and operation time in quantitative performance, and raises several technical solutions proposed for physical miniaturization and circuit control. In aspect of geometric layout, engineer team compress the character line thickness to suppress column height and reorder the source contact and isolation trench structure. The arrangement of horizontal memory holes approaches the geometric fill rate limit of the hexagonal densest arrangement from a loose matrix. Maximize areal density without reducing unit size. In circuit structure layer, the CMOS logic and register circuits, which were originally arranged side-by-side at the edge of the array, were moved to the area directly below the storage array. It not only decreases the area of grain but also allows putting more independent planes to extend the writing parallelism in chip. Meanwhile, place the character-driven circuit in the middle of local character line, the load capacitance and resistance of the long wire are halved to suppress charging and discharging delay. Aim to the short-term leakage, the industry has adopted a tiered coarse-fine write solution, scheme 16-16 first roughly outlines 16 relaxed states, and then performs fine-tuning to narrow boundaries after initial leakage current settlement. The 8-16 scheme divides the first phase into only 8 states, thereby reducing the number of verifications and shortening the writing time. In order to solve the issue of giant block, hierarchy block scheme divides a single vertical column into multiple independent segments. During erasure, a 20V high voltage is applied to the entire channel. The word lines of the unselected segments are simultaneously pushed up to 20V to suppress tunneling at the same potential, thus successfully erasing small, logical blocks without destroying the physical word string.

4. Pros and Cons

   For pros aspect, the structure of paper is highly systematic. It clearly constructs a 3D address of 3D NAND and operation model. Describe how the BE-ONO layer could erase through electric field tilt to fulfill DT by energy band theory. Keep

completely 3-layer physical thickness to block off the physical conflicts in standby. At the same time, it objectively describes the microcosmic physical defects of charge trapping unit. The short-term storage degradation would decompose to lateral migration, vertical relaxation, trap-assisted tunneling, and tunneling layer gate detachment, providing a solid microcosmic physical basis for the design of subsequent write algorithms. In addition, the block-by-deck architecture breakthrough the traditional physical block restrictions of complete erasing, use the synchronous boost cleverly divide the independent erasing space in vertical column. It provides a clear engineering model for bottlenecks of large capacity SSD garbage collection.

On the other hand, cons aspect. Although it takes control circuit under the memory array to save space, the paper doesn't raise the huge cost by this method. After the implementation of control circuit, it must meet high-temperature heating manufacturing process in hundreds of levels of memory. It would lead to the below components of circuit deterioration and leakage by high temperature. It would even cause high-voltage component failure. Meanwhile, the paper makes a low rating for Xtacking, only make it as a secondary variation. In fact, Xtacking let separately manufacturing for control circuit and memory array, both could use the most suitable heat and process. Combine them after finishing the work. It deals with the issue of breaking circuit by high-temperature, and it releases the high-speed I/O for data.

5. Future Work in My Opinion

   For the design for circuit architecture and algorithms for 3D NAND, in my opinion, I think that it has some potential research sides that are worth exploring for meet the actual applications in the future.

   First one is for the control method. Instead of the traditional fixed standard, we could import the more flexible dynamic adjustment algorithm. The approaches which raised by the paper, the core of them is use the single fixed standard setting to do the operations. But it is sensitive in reality for flash memory, like chip heating or service life aging, it would let the internal practical electric charges escape at different speeds. For changing these conditions, if the controller still uses same method to do reading and writing, it will only let the fail rating of reading increase rapidly. Then it would waste more time and computing to retry and test in backend. I suppose that the future research work should add some light analysis logic in controller, tracking and measuring the times of operations and current temperature for every block. Adjusting the most suitable read and write depends on the situation then. This dynamic work could not only reduce the lag of reading but also save the affordability of checking repeatedly to extend the actual

lifespan of hard disks.